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C.E.O.A MAT. HR. SEC. SCHOOL MELUR

MARCH - 2024 physics

I. ONE mark answer key

1. c). 1.1 eV

2. c) 480 W

$P = VI \Rightarrow P = \frac{V \times V}{R} \Rightarrow P = \frac{240 \times 240}{120}$

$P = 480 W$

3. (a) $\frac{Q}{\sqrt{2}}$

$T.E = \frac{Q^2}{2C} \Rightarrow \frac{1}{2}\frac{Q^2}{C} = \frac{q^2}{2C} + \frac{LI^2}{2}$

$\therefore \frac{q^2}{2C} = \frac{LI^2}{2}$

$\frac{Q^2}{2C} = \frac{q^2}{2C} + \frac{q^2}{2C}$

$\frac{Q^2}{\cancel{2C}} = \frac{2q^2}{\cancel{2C}} \Rightarrow q^2 = \frac{Q^2}{2}$

$\boxed{q = \frac{Q}{\sqrt{2}}}$

4. (d) 3750 Å

$\phi = \frac{hc}{\lambda_0} \Rightarrow \phi = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{3.313 \times 1.6 \times 10^{-19}}$

$\phi = 3.750 \times 10^{-7} m$

(or)

$\phi = 3750 Å$

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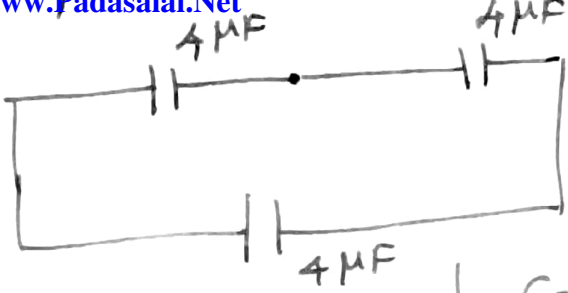
5. (d) 6 μF



$$\frac{1}{C_s} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}$$

$$\boxed{C_s = 2\,\mu F}$$

$$C_p = 2 + 4$$

$$\boxed{C_p = 6\,\mu F}$$

6. (a) photo voltaic action

7. (d) its wavelength

8. (c) 900 Vm^{-1}

$$v = \frac{E}{B} \Rightarrow E = v.B$$

$$E = 3\times10^{8} \times 3\times10^{-6} \Rightarrow E = 9\times10^{2}\ Vm^{-1}$$

(or)

$$E = 900\ Vm^{-1}$$

9. (d) $\frac{3}{\pi} P_m$

$$\theta = 60^\circ = \pi/3 \text{ rad}$$

$$l' = r \qquad M = P_m$$

$$M = m.l \qquad M' = m.l'$$

$$\frac{M}{M'} = \frac{l}{l'} \Rightarrow \frac{\pi}{3} = \frac{l}{r} \Rightarrow l = \frac{\pi r}{3}$$

$$\frac{M}{M'} = \frac{\frac{\pi r}{3}}{r} \Rightarrow \frac{M}{M'} = \frac{\pi \cancel{r}}{3} \times \frac{1}{\cancel{r}}$$

$$\frac{M}{M'} = \frac{\pi}{3} \Rightarrow M' = \frac{3M}{\pi} \text{ (or) } M' = \frac{3}{\pi} P_m$$

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10). (b) $\frac{\pi}{4}$

$R = X_L$ $\tan\phi = \frac{X_L}{R} = 1$

$\tan\theta = 1 \Rightarrow \theta = \tan^{-1}(1)$

$\theta = 45^\circ$ (or) $\frac{\pi}{4}$

11. (a) more than before

12. (a) 3

$I.E = 122.4\,V$ $n = 1$

$Z = ?$

$I.E = \frac{13.6\,Z^2}{n^2}$

$Z^2 = \frac{I.E \times n^2}{13.6} \Rightarrow Z^2 = \frac{122.4 \times 1^2}{13.6}$

$Z^2 = 9 \Rightarrow \boxed{Z = 3}$

13. (c) polarisation

14. (a) plane polarised

15. (a) Albert Einstein

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